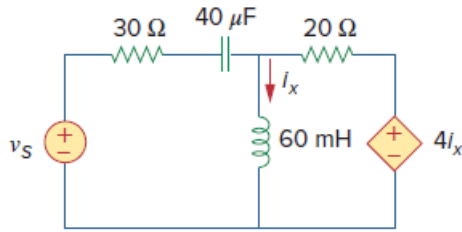


Problem

Find the complex power delivered by v_s to the network in Fig. 11.69. Let $v_s = 100 \cos 2000t$ V.

Figure 11.69:



Step-by-step solution

Step 1 of 5

Refer to Figure 11.69 in the textbook.

The source voltage is,

$$v_s = 100 \cos(2000t) \text{ V}$$

From the source voltage, the frequency is, $\omega = 2000 \text{ rad/s}$.

Convert the voltage source into phasor.

$$V_s = 100 \angle 0^\circ \text{ V}$$

Calculate the impedance of the capacitor.

$$\begin{aligned} Z_c &= \frac{1}{j\omega C} \\ &= \frac{1}{j(2000)(40 \times 10^{-6})} \\ &= -j12.5 \Omega \end{aligned}$$

Calculate the impedance of the inductor.

$$\begin{aligned} Z_L &= j\omega L \\ &= j(2000)(60 \times 10^{-3}) \\ &= j120 \Omega \end{aligned}$$

Step 2 of 5

Draw the equivalent frequency domain circuit.

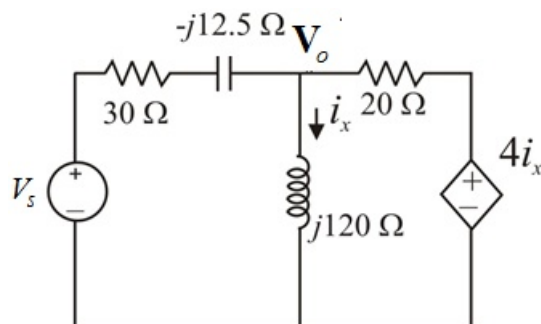


Figure 1

Step 3 of 5

From the circuit,

$$I_x = \frac{V_o}{j120}$$

Write the Kirchhoff's current law equation at node v_o .

$$\frac{V_o - 100\angle 0^\circ}{30 - j12.5} + \frac{V_o - 4I_x}{20} + \frac{V_o}{j120} = 0$$

Substitute I_x for $\frac{V_o}{j120}$.

$$\frac{V_o - 100\angle 0^\circ}{30 - j12.5} + \frac{V_o - 4\left(\frac{V_o}{j120}\right)}{20} + \frac{V_o}{j120} = 0$$

$$\left[\frac{1}{30 - j12.5} + \frac{1 - \frac{4}{j120}}{20} + \frac{1}{j120} \right] V_o = \frac{100\angle 0^\circ}{30 - j12.5}$$

$$\left[0.0284 + j0.0118 + \frac{1 + j0.033}{20} - j0.00833 \right] V_o = 2.84 + j1.183$$

$$[0.0784 + j0.00512] V_o = 2.84 + j1.183$$

$$V_o = 37.05 + j12.7 \text{ V}$$

$$V_o = 39.15\angle 18.87^\circ \text{ V}$$

Step 4 of 5

Calculate the current through source.

$$\begin{aligned} I_o &= \frac{V_s - V_o}{30 - j12.5} \\ &= \frac{100 - (39.15\angle 18.87^\circ)}{30 - j12.5} \\ &= 1.937 + j0.385 \text{ A} \end{aligned}$$

Step 5 of 5

Calculate the complex power, S delivered by the source.

$$\begin{aligned} S &= \frac{1}{2} V_s I_o^* \\ &= \frac{1}{2} (100)(1.937 - j0.385) \\ &= 96.85 - j19.25 \text{ VA} \end{aligned}$$

Therefore, the complex power, S delivered by the source is $\boxed{96.85 - j19.25 \text{ VA}}$.